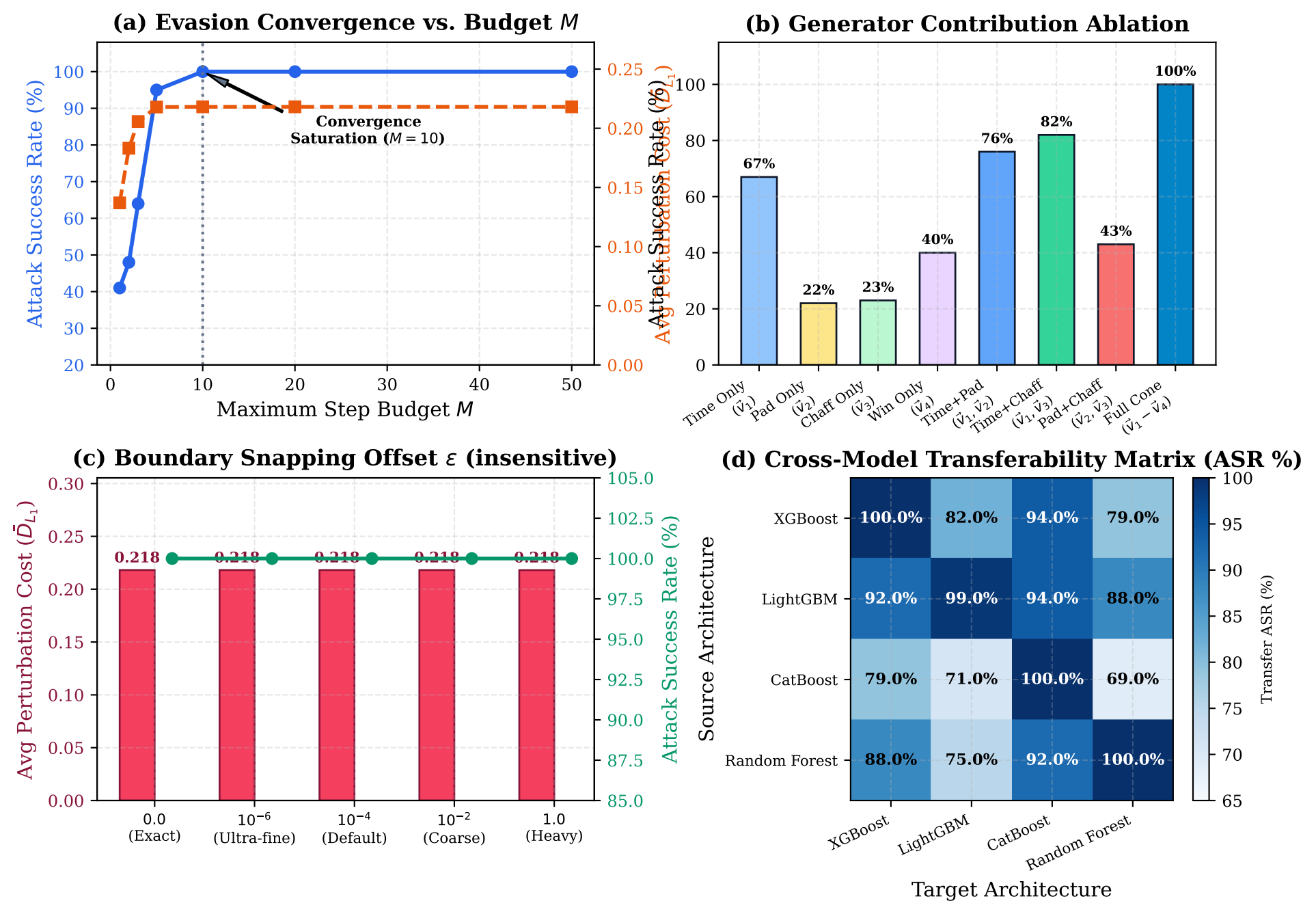
(a) Evasion Convergence vs. Budget M
Attack Success Rate (%)
Avg Perturbation Cost ($\bar{D}_{L_1}$)
Maximum Step Budget M
Convergence Saturation ($M = 10$)
(b) Generator Contribution Ablation
Attack Success Rate (%)
67%
22%
23%
40%
76%
82%
43%
100%
Time Only ($\vec{v}_1$)
Pad Only ($\vec{v}_2$)
Chaff Only ($\vec{v}_3$)
Win Only ($\vec{v}_4$)
Time+Pad ($\vec{v}_1, \vec{v}_2$)
Time+Chaff ($\vec{v}_1, \vec{v}_3$)
Pad+Chaff ($\vec{v}_2, \vec{v}_3$)
Full Cone ($\vec{v}_1 - \vec{v}_4$)
(c) Boundary Snapping Offset ε (insensitive)
Avg Perturbation Cost ($\bar{D}_{L_1}$)
Attack Success Rate (%)
0.218
0.0 (Exact)
10^{-6} (Ultra-fine)
10^{-4} (Default)
10^{-2} (Coarse)
1.0 (Heavy)
(d) Cross-Model Transferability Matrix (ASR %)
Source Architecture
Target Architecture
Transfer ASR (%)
XGBoost
LightGBM
CatBoost
Random Forest
100.0% 82.0% 94.0% 79.0%
92.0% 99.0% 94.0% 88.0%
79.0% 71.0% 100.0% 69.0%
88.0% 75.0% 92.0% 100.0%